Practice Problems 03 Solutions

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QUESTION 01

The relationship between school funding and student performance is explored in the following data, which shows per-pupil expenditures (X, in \$1000s) and graduation rate (Y, in %) for 26 randomly chosen districts in Massachusetts. The sums are calculated as follows:

$\sum_{i=1}^{26} x_i = 360$ $\quad \sum_{i=1}^{26} y_i = 2256.6$ $\quad \sum_{i=1}^{26} x_i^2 = 5365.08$ $\quad \sum_{i=1}^{26} x_i y_i = 31402$ $\quad \frac{s}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2}} = 0.6044$

a) Calculate the least squares regression line, listing your estimates for the slope and intercept.

```r
n <- 26
sum.x <- 360
sum.y <- 2256.6
sum.x2 <- 5365.08
sum.xy <- 31402
```

$b = \frac{n\sum_{i=1}^{n} x_i y_i - (\sum_{i=1}^{n} x_i)(\sum_{i=1}^{n} y_i)}{n(\sum_{i=1}^{n} x_i^2) - (\sum_{i=1}^{n} x_i)^2}$

$a = \frac{\sum_{i=1}^{n} y_i - b\sum_{i=1}^{n} x_i}{n} = \bar{y} - b\bar{x}$

```r
(b <- ( n*sum.xy - (sum.x*sum.y) )/ ((n*sum.x2) - sum.x^2)) # slope
```

```
## [1] 0.4120468
```

```r
(a <- (sum.y-b*sum.x)/n) # intercept
```

```
## [1] 81.08704
```

b) Perform a hypothesis test with significance level 0.05 to determine whether the slope is non-zero. List hypotheses, test statistic, and p-value.

$H_0 : \beta_1 = 0$ $H_1 : \beta_1 \neq 0$

```
(t <- (b - 0)/0.6044) # test statistic--divisor is the last provided value: s/sqrt(sum((x-mean(x))^2))
```

```
## [1] 0.6817452
```

```
(p.val <- 2*(1-pt(abs(t),n-2))) # p-value
```

```
## [1] 0.5019266
```

c) Construct a 95% confidence interval for the slope.

```
alpha <- 0.05
t.crit <- qt(1-alpha/2,n-2) # critical value (number of steps)
se <- 0.6044 # error term (size of steps)
ME <- t.crit*se # margin of error

c(b-ME,b+ME) # Confidence Interal
```

```
## [1] -0.8353735  1.6594671
```

d) What do you conclude about the data (in context) based on the hypothesis test and confidence interval you found?

We fail to reject H_0, and our confidence interval contains 0. This implies the slope is not signficantly different from zero, and thus spending per student is not an effective predictor of graduation rate.

QUESTION 02

A candy store has a bin with three pounds of M&Ms in various colors. Supposedly, there are three frequencies associated with the colors: 30% are brown, 20% each are yellow and red, and 10% each are orange, blue, and green. Test at significance level 0.05 the hypothesis that these frequencies are consistent with the counts from the three-pound bin below:

Colors	Number
Brown	455
Yellow	343
Red	318
Orange	152
Blue	130
Green	129

a) List the null and alternative hypotheses for this test. H_0: The candy proportions follow the given probabilities. H_1: The candy proportions do NOT follow the given probabilities.

b) Calculate the expected counts for each color.

```
k <- 6 # six categories of candy color
obs <- c(455,343,318,152,130,129) # observed counts
n <- sum(obs) # total sample size
p.exp <- c(0.3,0.2,0.2,0.1,0.1,0.1) # expected proportions
(exp <- n*p.exp) # expected counts, given total sample size n
```

```
## [1] 458.1 305.4 305.4 152.7 152.7 152.7
```

c) Calculate the test statistic and its corresponding p-value.

```
(chi.test <- sum( (obs-exp)^2/exp) )# test statistic
```

```
## [1] 12.22615
```

```
(p.val <- 1-pchisq(chi.test, k-1)) # p-value
```

```
## [1] 0.031817
```

d) What can you conclude about the color frequencies based on the results of your hypothesis test?

We reject H_0 and conclude that the candy colors do NOT match the provided proportions (e.g. at least one proportion is different from expected).

QUESTION 03

Mount Etna erupted in 1669, 1780, and 1865. When molten lava hardens, it retains the direction of the Earth's magnetic field. Three blocks of lava were examined from each of these eruptions and the declination of the magnetic field in the block was measured. The results are given in the following table. Do these data suggest that the direction of the Earth's magnetic field shifted over the time period spanned by the eruptions? Use significance level 0.05.

1669	1780	1865
57.8	57.9	52.7
60.2	55.2	53.0
60.3	54.8	49.4

Setting up data

```r
k <- 3 # 3 groups
r <- 3 # 3 observations per group
n <- k*r # 9 total observations

y.1669 <- c(57.8,60.2,60.3)
y.1780 <- c(57.9,55.2,54.8)
y.1865 <- c(52.7,53.0,49.4)
etna <- data.frame(y.1669,y.1780,y.1865)

means <- colMeans(etna) # group means
mean.tot <- mean(means) # overall mean
```

Calculating SSG

```r
( SSG <- sum(r*sum( (means-mean.tot)^2 )) )
```

```
## [1] 90.02667
```

Calculating SSE

```r
sse.parts <- rep(NA,r)

# for each group...
for(i in 1:k){
  sse.parts[i] = sum( (etna[,i] - mean(etna[,i]))^2 )
}

( SSE <- sum(sse.parts) )
```

```
## [1] 17.67333
```

Calculating F statistic and p-value

```r
MSG <- SSG/(k-1)
MSE <- SSE/(n-k)

(F.stat <- MSG/MSE) # test statistic
```

```
## [1] 15.28178
```

```r
(p.val <- 1-pf(F.stat,k-1,n-k))
```

```
## [1] 0.00441884
```

Since our p-value is less than $alpha = 0.05$, we REJECT H_0 and conclude that not all years have equal average declinations. The magnetic field does appear to change over time.